

26春实变函数期中考试 孙红蕊

1, 辨析题

(1) $E_k \subset \mathbb{R}^n$, E_k has no limit point, then $\bigcup_{k \geq 1} E_k$ has no limit point.

eg.1: $E_k = \{\frac{1}{k}\}$, $\bigcup_{k \geq 1} E_k = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$, then it has limit point 0.

eg.2: write $\mathbb{Q} \cap [0, 1] = \{r_k\}$, let $E_k = \{r_1, \dots, r_k\}$,

then E_k has no limit point, but $\bigcup_{k \geq 1} E_k = \mathbb{Q} \cap [0, 1]$ has limit point.

(2) $G \subset \mathbb{R}^n$ is open, then $|G| = |\bar{G}|$.

$\bar{G} = G \cup \partial G$ and $|\partial G| = 0$.

(3) $A \subset B \subset C$, $A \sim C$ then $A \sim B$.

use the Bernstein Thm.

(4) $f \in C(F_1)$, $f \in C(F_2)$, F_1 and F_2 are closed $F_1 \cap F_2 = \emptyset$, then $f \in C(F_1 \cup F_2)$.

(5) $f_k \xrightarrow{u} 0$ in E , then $\int_E f_k \rightarrow 0$.

NOT True.

$$f_k = \begin{cases} \frac{1}{k}, 0 \leq x \leq k \\ 0, x > k \end{cases}$$

2, let $f(x, y) = \frac{x^4 - y^4}{x^4 + y^4}$, $x, y \in (0, 1]^2$, Compute the $(L) \int_{(0,1]^2} f(x, y) dx dy$.

we need to show the Lebesgue integral can be converted to repeated integral,

then compute the Riemann integral.

Rem: $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ 萃英考试题

3, 作业原题: Let $\{E_k\}$ are measurable sets in $\mathbb{R}^n$, and $\sum_{k \geq 1} |E_k| < \infty$, show that $|\overline{\lim}_{k \rightarrow \infty} E_k| = 0$.

4, $\{E_k\}$ are measurable sets in $\mathbb{R}^n$, $\{E_k\}$ are uniformly bounded in $\mathbb{R}^n$, $\lim_{k \rightarrow \infty} E_k = E$.

show that $\lim_{k \rightarrow \infty} |E_k| = |E|$.

Solution:

let $f_k = \chi_{E_k}$, $f = \chi_E$, defined in $\mathbb{R}^n$, then use the Fatous' Lemma and generalized LDCT(作业题)

5, (Lusin Thm) state the relation between continuous function and measurable function defined in $E \subset \mathbb{R}^n$ (E is measurable), and prove it.

6, 作业原题: If $f_k \xrightarrow{m} f$ in E , $|f_k| \leq \varphi$ a.e., $\varphi \in L(E)$, show that $f \in L(E)$ and $\int_E |f_k - f| \rightarrow 0$.

Solution:

the proof of $f \in L(E)$:

$f_k \xrightarrow{m} f \implies \exists \{f_{k_l}\}, f_{k_l} \rightarrow f$ a.e.,

$|f_k| \leq \varphi$ a.e.

$\implies |f_{k_l}| \leq \varphi$ a.e.

$$\implies |f| \leq \varphi \text{ a.e.}$$

$$\implies |f| \in L(E) \implies f \in L(E).$$

the proof of $\int_E |f_k - f| \rightarrow 0$:

It's easy to know that $f_k \in L(E)$, by $f \in L(E)$, It suffices to show that $\int_E f_k \rightarrow \int_E f$.

$$f_k \xrightarrow{m} f \implies f_{k_i} \xrightarrow{m} f \text{ for any subsequence } \{f_{k_i}\} \text{ of } \{f_k\},$$

By Riesz Thm, $\exists$ a subsequence $\{f_{k_{i_j}}\}$ of $\{f_{k_i}\}$ s.t. $f_{k_{i_j}} \rightarrow f$ a.e. in E ,

$$\text{By LDCT, } \int_E f_{k_{i_j}} \rightarrow \int_E f.$$

Since $\{\int_E f_{k_i}\}$ is a subsequence of $\{\int_E f_k\}$, $\{\int_E f_{k_{i_j}}\}$ is a subsequence of $\{\int_E f_{k_i}\}$,

Now we obtain for any subsequence of $\{\int_E f_{k_i}\}$ of $\{\int_E f_k\}$, it has a subsequence $\{\int_E f_{k_{i_j}}\}$ that converges to $\int_E f$.

By the property of real number sequences, we get $\int_E f_k \rightarrow \int_E f$.

Another Solution: use the Absolute Continuity of Integral.

萃英试题:

兰州大学萃英学院考试试卷

2020 学年第 学期

萃英院 专业 史英奇

考生姓名 郭晓东 准考证号 3202301 年级 课程

阅卷人签名

题号	一	二	三	四	五	总分
分数						
阅卷教师						

1. (i) $\{E_k\} \subset \mathcal{R}$, $\mathcal{R}(\bigcup_{k=1}^{\infty} E_k) = \bigcup_{k=1}^{\infty} \mathcal{R}(E_k)$
(ii) A, B, C 不相交, $A \subset B \subset C$ if $A \sim C$ 则 $A \sim B$
(iii) $G \subset \mathcal{R}$ 则 $|G| = |\mathcal{R}|$
(iv) P_1, P_2 在 $\mathcal{R}$ 中, $P_1 \cap P_2 = \emptyset$, if $f: P_1 \cup P_2 \rightarrow \mathcal{R}$
 $f \in \mathcal{C}(P_1)$ if $f \in \mathcal{C}(P_2)$
则 $f \in \mathcal{C}(P_1 \cup P_2)$
(v) $\{f_k\} \subset L(E)$ $\int_E f_k d\mu \rightarrow 0$ 则 $f_k \rightarrow 0$ a.e. in E
2. $f(x,y) = \frac{x^2+y^2}{x^2+y^2+1}$ $(x,y) \in (0,1)^2$ 计算 $\int_{(0,1)^2} f(x,y) dxdy$

2. $\{E_k\}_{k=1}^{\infty}$ 为可测集, $\sum_{k=1}^{\infty} \mu(E_k) < \infty$ $\mu(\bigcap_{k=1}^{\infty} E_k) = 0$
4. $\{E_k\}$ 在 $\mathcal{R}$ 中, 可测集 $E \subset \mathcal{R}$ $\mu(E) < \infty$
且 $E_k \subset E$ $\bigcup_{k=1}^{\infty} E_k = E$ 证: $\lim_{k \rightarrow \infty} \mu(E_k) = \mu(E)$
5. $P \subset [a,b]$ 为可测集 $\mu(P) = 0$ 证: X_P 在 $[a,b]$ 上黎曼可积
6. $E \subset \mathcal{R}$ 可测, 证: 可测集与连续函数关系, 并证明可测集 E
7. $\{f_k\}$ 在 E 上可测, 证: $\lim_{k \rightarrow \infty} f_k = f$ a.e. in E
证: $f \in L(E)$ 且 $|f_k| \leq f$ a.e. in E
证: $f \in L(E)$ and $\lim_{k \rightarrow \infty} \int_E |f_k - f| d\mu = 0$

作题区
(1) 错误
(i) $\forall x \in (\bigcup_{k=1}^{\infty} E_k)$
则 $\exists k_0$ s.t. $x \in E_{k_0}$ 则 x 也属于某个 (E_k)
即: $\exists k_0$ s.t. $x \in E_{k_0}$ 则 $\lim_{k \rightarrow \infty} x_k = x$
容易分析: 上述推理成立的前提是: 任意 $x \in (\bigcup_{k=1}^{\infty} E_k)$ $(\leq \infty)$
且 $0(x, \infty) \subseteq$ 某个 E_k , 即 x 是某个 E_k 的聚点
但 $\{E_k\}$ 的性质未知, 若取每个 E_k 为离散点集或孤立点集, 则上述推理不成立.
第2页 (共8页)